Day 03: Network Interfaces

Muhammad Usman
Cybersecurity Learning Log

Part 1: Theory Questions and Answers

1. What is a Network Interface Card (NIC)?

A Network Interface Card (NIC) is a hardware component that allows a computer to connect to a network. It may provide a wired Ethernet connection or a wireless Wi-Fi connection.

2. What is Ethernet?

Ethernet is a widely used technology for connecting devices in a wired local area network (LAN). It defines how data is transmitted between connected devices.

3. What is RJ-45?

RJ-45 commonly refers to the eight-position connector used with Ethernet network cables. It plugs into compatible Ethernet ports on computers, switches, and routers.

4. What does UTP stand for?

UTP stands for Unshielded Twisted Pair. It is a copper cable containing pairs of insulated wires twisted together to help reduce electrical interference.

5. Name three common Ethernet cable categories.

Cat5e, Cat6, and Cat6a are three common Ethernet cable categories.

6. What is a straight-through cable?

A straight-through Ethernet cable uses the same wiring standard at both ends, such as T568A at both ends or T568B at both ends. It is commonly used to connect different types of network devices, such as a computer to a switch.

7. What is a crossover cable?

A crossover cable traditionally uses T568A at one end and T568B at the other. It was commonly used to connect similar Ethernet devices directly.

8. Explain Auto MDI-X.

Auto MDI-X automatically detects and adjusts the transmit and receive pairs needed for an Ethernet connection. On compatible devices, it removes the need to select a straight-through or crossover cable manually.

9. Difference between copper and fiber-optic networking?

Copper networking carries electrical signals through metal wires. Fiber-optic networking transmits data as light through glass or plastic fibers. Fiber supports long distances and resists electromagnetic interference; copper is often less expensive and can support Power over Ethernet.

10. What is multimode fiber?

Multimode fiber has a relatively large core that allows multiple light paths, or modes, to travel through it. It is commonly used for shorter-distance connections, such as within buildings and data centers.

11. What is single-mode fiber?

Single-mode fiber has a very small core designed to carry light primarily along one mode. It is suitable for high-bandwidth connections over long distances.

12. Two advantages of fiber-optic networking.

First, fiber supports high bandwidth over long distances. Second, it is resistant to electromagnetic interference and does not conduct electrical current through the fiber itself.

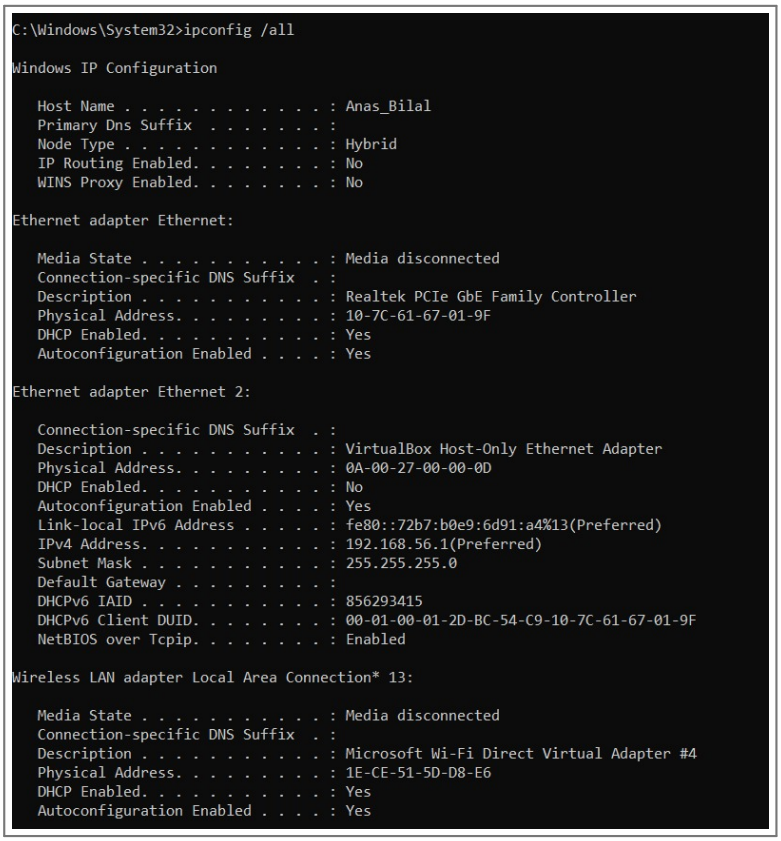
13. Why is understanding network interfaces important for cybersecurity?

It helps analysts identify how devices connect to networks, understand IP and MAC addresses, troubleshoot connectivity, investigate suspicious activity, and recognize unexpected interfaces or network exposure.

Part 2: Practical Lab (Windows ipconfig /all)

The following screenshots were taken on a Windows computer using the **ipconfig /all** command. They show every network interface on the machine and the IP configuration of the active one.

Figure 1: ipconfig /all - Host Details and Ethernet Adapters



```
C:\Windows\System32>ipconfig /all

Windows IP Configuration

   Host Name . . . . . . . . . . . . : Anas_Bilal
   Primary Dns Suffix  . . . . . . . :
   Node Type . . . . . . . . . . . . : Hybrid
   IP Routing Enabled. . . . . . . . : No
   WINS Proxy Enabled. . . . . . . . : No

Ethernet adapter Ethernet:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Realtek PCIe GbE Family Controller
   Physical Address. . . . . . . . . : 10-7C-61-67-01-9F
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 2:

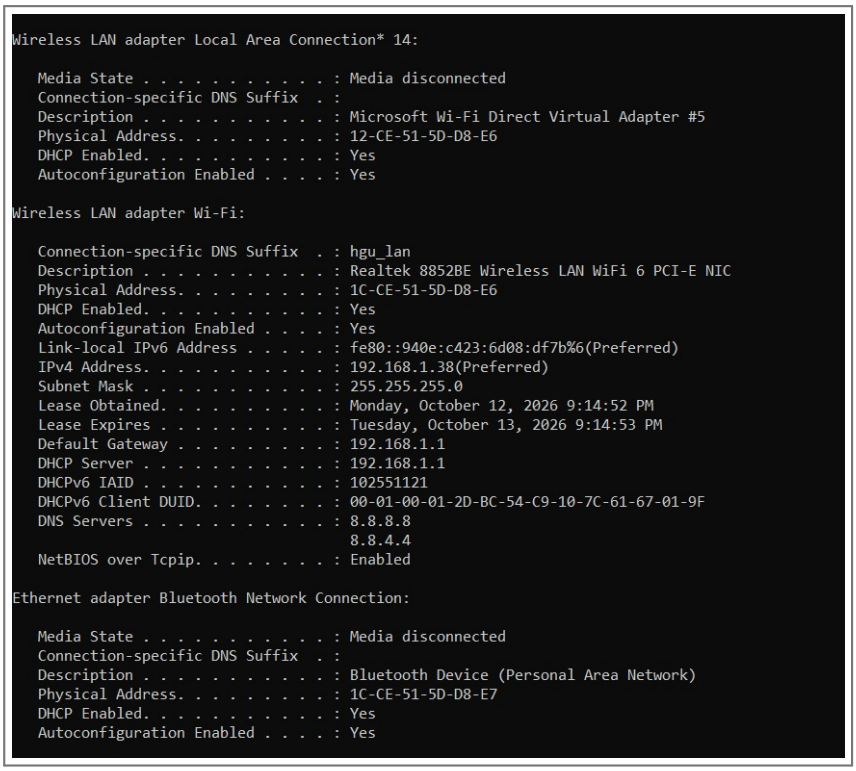
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : VirtualBox Host-Only Ethernet Adapter
   Physical Address. . . . . . . . . : 0A-00-27-00-00-0D
   DHCP Enabled. . . . . . . . . . . : No
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::72b7:b0e9:6d91:a4%13(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.56.1(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
   DHCPv6 IAID . . . . . . . . . . . : 856293415
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-BC-54-C9-10-7C-61-67-01-9F
   NetBIOS over Tcpip. . . . . . . . : Enabled

Wireless LAN adapter Local Area Connection* 13:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #4
   Physical Address. . . . . . . . . : 1E-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

The command lists all network adapters. **Ethernet** (Realtek PCIe GbE Family Controller) shows *Media disconnected*, so no cable is plugged in. **Ethernet 2** is a VirtualBox Host-Only adapter with the IPv4 address **192.168.56.1** and subnet mask 255.255.255.0. It has DHCP disabled and no default gateway. **Local Area Connection* 13** is a Microsoft Wi-Fi Direct Virtual Adapter and is also disconnected.

Figure 2: ipconfig /all - Active Wi-Fi Adapter



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Wireless LAN adapter Local Area Connection* 14:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #5
   Physical Address. . . . . . . . . : 12-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : hgu_lan
   Description . . . . . . . . . . . : Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC
   Physical Address. . . . . . . . . : 1C-CE-51-5D-D8-E6
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
   Link-local IPv6 Address . . . . . : fe80::940e:c423:6d08:df7b%6(Preferred)
   IPv4 Address. . . . . . . . . . . : 192.168.1.38(Preferred)
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Lease Obtained. . . . . . . . . . : Monday, October 12, 2026 9:14:52 PM
   Lease Expires . . . . . . . . . . : Tuesday, October 13, 2026 9:14:53 PM
   Default Gateway . . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . . : 192.168.1.1
   DHCPv6 IAID . . . . . . . . . . . : 102551121
   DHCPv6 Client DUID. . . . . . . . : 00-01-00-01-2D-BC-54-C9-10-7C-61-67-01-9F
   DNS Servers . . . . . . . . . . . : 8.8.8.8
                                       8.8.4.4
   NetBIOS over Tcpip. . . . . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :
   Description . . . . . . . . . . . : Bluetooth Device (Personal Area Network)
   Physical Address. . . . . . . . . : 1C-CE-51-5D-D8-E7
   DHCP Enabled. . . . . . . . . . . : Yes
   Autoconfiguration Enabled . . . . : Yes
```

The **Wi-Fi** adapter (Realtek 8852BE Wireless LAN WiFi 6 PCI-E NIC) is the active interface. It received the IPv4 address **192.168.1.38** with subnet mask 255.255.255.0 from the DHCP server **192.168.1.1**. The default gateway is **192.168.1.1** and the DNS servers are **8.8.8.8** and **8.8.4.4**. The Bluetooth Network Connection and Wi-Fi Direct adapter (Local Area Connection* 14) show *Media disconnected*.

Part 3: IP Address and DNS Server Summary

Item	Value	Source
Active interface	Wi-Fi (Realtek 8852BE WiFi 6)	Figure 2
IPv4 address	192.168.1.38	Figure 2
Subnet mask	255.255.255.0	Figure 2
Default gateway	192.168.1.1	Figure 2
DHCP server	192.168.1.1	Figure 2
DNS servers	8.8.8.8 (primary), 8.8.4.4 (secondary)	Figure 2
MAC address (Wi-Fi)	1C-CE-51-5D-D8-E6	Figure 2
VirtualBox Host-Only IPv4	192.168.56.1	Figure 1

IP address: identifies a network interface so that devices can send and receive packets. This computer uses 192.168.1.38 on its Wi-Fi interface and reaches other networks through the gateway 192.168.1.1.

DNS server: the server the computer asks to translate domain names (such as example.com) into IP addresses. Here the DNS servers are 8.8.8.8 and 8.8.4.4.